

TANISHK JAIN

Machine Learning | Full-Stack

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Summary

B.Tech Computer Science (AI & ML) student with experience across machine learning, computer vision, and full stack development. Built end-to-end deep learning pipelines for medical image segmentation and deployed production web applications.

Education

Vellore Institute of Technology

2023 – 2027

B.Tech in Computer Science Engineering (AI & ML), CGPA: 8.34 / 10.0

Technical Skills

Languages: Python, C++, Java, Javascript

AI / ML: TensorFlow, Scikit-Learn, OpenCV, MediaPipe; Deep Learning, Computer Vision, Image Segmentation, Prompt Engineering

Libraries: NumPy, Pandas, Matplotlib, SHAP, LIME

Full Stack: NextJS, AngularJS, Django REST Framework, Supabase, SQL

Agentic AI: Claude Code, GitHub Copilot, Antigravity

Tools: Git, GitHub, Jupyter, Google Colab, VS Code

Experience

Indian Institute of Technology Delhi

June 2026 – September 2026

Machine Learning Research Intern (On-site)

Delhi, India

- Developed deep learning pipelines for retinal vessel segmentation and artery/vein classification under Prof. Tapan Kumar Gandhi.
- Trained and benchmarked encoder-decoder segmentation architectures including Attention U-Net, tuning loss functions and hyperparameters against IoU and Dice coefficient.

DJUBO

Jan 2026 – Apr 2026

Software Engineer Intern (Hybrid) | GitHub

Noida, India

- Built a large-scale employee attendance management platform using AngularJS, Django REST Framework, SQLite, and token-based authentication.
- Implemented REST endpoints powering dashboards, calendar views, admin workflows, leave management, and analytics visualizations via Handsontable and Highcharts.

Projects

Parkinson's Detection from Voice with Explainable AI | *Python, Scikit-Learn, SHAP, LIME*

GitHub

- Identified speaker leakage in conventional train-test splits, showing models were learning speaker identity rather than disease characteristics.
- Evaluated 12 models across 960 experimental runs, showing conventional evaluation inflated ROC-AUC by 0.159 on average and changed model rankings.
- Applied SHAP and LIME on unseen speakers, achieving 0.924 ROC-AUC with a reduced two-feature screening model at 92% sensitivity.

Real-Time Attention Monitoring System | *Python, OpenCV, MediaPipe*

GitHub

- Built a real-time webcam system that analyzes facial landmarks, head pose, and posture to detect distraction.
- Combined these signals in a rule-based scoring engine to monitor attention frame by frame.

Hisaab: Group Trip Expense Manager | *React, TypeScript, Supabase*

GitHub | Live

- Built and deployed a group expense manager supporting optimized settlements, uneven splits, and multi-payer transactions.
- Implemented offline-first storage with background Postgres sync and row-level security.

Certifications

Applied Machine Learning, University of Michigan | AWS Technical Essentials | ServiceNow Certified System Administrator (CSA)

Leadership & Achievements

Microsoft Technical Club, VIT Bhopal

Dec 2024 – May 2026

Vice President (Member → Team Co-Lead → Team Lead)

Bhopal, India

- Planned and executed a complete restructuring of 75+ club members, overseeing new-member interviews, team reallocation, and selective member transitions to strengthen team composition and operations.
- Led multiple events, including two flagship AdvITya 2026 events across technical and non-technical tracks, coordinating multiple teams and owning end-to-end execution during my tenure.

VIT Bhopal x Johns Hopkins University HealthHack

February 2026

Team Leader, Finalist | Repository

Bhopal, India

- Led a team of 6 to build CareFast, a mobile-first emergency healthcare application, outperforming 150+ teams to advance to the final round.